

DELIVERABLE 3: Use-Case Flow Specification (1-Page Formal Document)

Problem Statement #55: Book Club Reading Challenge & Discussion Portal | Author: Sushruth S (PES1UG24CS486)

Use Case ID:	UC-04	Use Case Name:	Vote in Monthly Book Selection Poll
Primary Actor:	Club Member	Secondary Actors:	Auth Service, Real-Time Tally Engine
Trigger:	Member opens active monthly book poll	Stereotypes:	«include» Session Verification, «include» Real-Time Tally
Preconditions: 1. Member must hold an authenticated, active account session on the portal. 2. An active monthly book selection poll must be open for voting. 3. Member has not yet submitted a vote in the current voting cycle.		Postconditions: 1. Member's single vote is recorded and permanently bound to their profile. 2. Aggregated poll tallies and percentages update across all live sessions. 3. Member UI transitions to read-only confirmation state with disabled buttons.	

Main Success Scenario (MSS):

- Navigate to Poll:** Club Member navigates to the 'Monthly Book Poll' section on the community dashboard.
- Retrieve Details:** System fetches active nominees, book synopsis previews, and verifies the member's voting eligibility.
- Display Options:** System displays nominated titles with interactive radio selection elements and current poll deadline.
- Select Book:** Club Member selects their desired book candidate from the list.
- Submit Ballot:** Club Member clicks the 'Submit Vote' button.
- Confirm Intent:** System presents a confirmation modal: *'Confirm your vote for [Selected Book]? This cannot be changed.'*
- Acknowledge:** Club Member confirms their selection.
- Validate & Lock:** System verifies the session token and initiates an atomic write transaction to record the ballot.
- Update Tallies:** System increments candidate score in Redis cache and pushes real-time websocket broadcast to active users.
- Render Success State:** System presents a 'Vote Recorded Successfully' receipt, reveals live graphical percentage distribution, and locks member voting controls.
- Conclusion:** Use case terminates successfully.

Alternate / Exceptional Flows:

3a. Member Has Already Voted in Active Cycle:

- 3a1. System detects existing vote timestamp in the member's profile for the active poll ID.
- 3a2. System displays current live results with notification: *'You have already voted in this cycle.'*
- 3a3. Voting controls remain permanently disabled. Use case terminates.

5a. Submission Without Selecting a Candidate:

- 5a1. System client-side validation detects empty selection upon clicking 'Submit Vote'.
- 5a2. System highlights the list in red with prompt: *'Please select one book before submitting.'*
- 5a3. Member selects a valid candidate and resumes at Step 5.

8a. Concurrent Session / Duplicate Vote Collision:

- 8a1. System database unique key constraint (UserID + PollID) detects a race-condition duplicate submission.
- 8a2. System rejects redundant payload, rolls back transaction, and returns HTTP 409 Conflict.
- 8a3. System displays alert: *'Duplicate vote prevented. Your previous vote is recorded.'*
- 8a4. System re-renders read-only results dashboard. Use case terminates.

8b. Poll Closes During User Interaction:

- 8b1. System detects current server timestamp is greater than poll expiration timestamp.
- 8b2. System rejects vote with notice: *'The voting window has closed. Thank you for participating.'*
- 8b3. System transitions UI to final closed results leaderboard. Use case terminates.